# Paper Mill Load Optimization: From Price Discovery to Implementation

## A Data-Driven Analysis of the NZ Electricity Market

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## The Central Question

Can a paper mill reduce electricity costs by shifting load based on price patterns?

This document answers that question through three stages:

1. Understanding price patterns in the NZ electricity market
2. Identifying operational flexibility in paper manufacturing
3. Designing a practical optimization system

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## Part 1: Price Pattern Discovery

### The Market Context

New Zealand's wholesale electricity market operates on 30-minute trading periods. Generators submit bids (offers to supply), and the market clears at a nodal price set by the marginal generator.

We analyzed 474,337 generator bid records across 15 days spanning March 2025 to March 2026 to understand price patterns.

**Data limitation:** These are bids (what generators offer), not clearing prices (what buyers pay). The clearing price equals the highest accepted bid. Patterns should hold, but absolute prices may differ.

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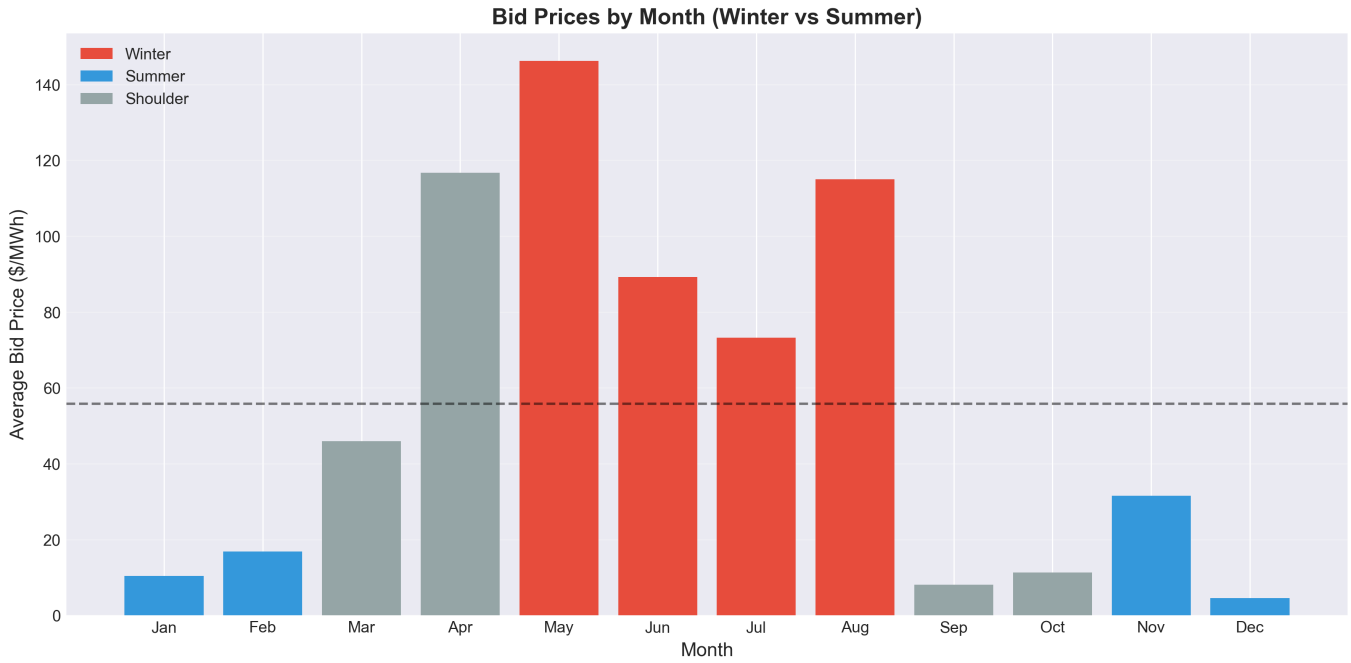
### Pattern 1: Season Dominates

Winter is 6.7 times more expensive than summer.




| Season | Average | Peak Month | Low Month |
|---|---|---|---|
| Winter (May-Aug) | $105.90/MWh | May: $146/MWh | Jul: $73/MWh |
| Summer (Nov-Feb) | $15.84/MWh | Nov: $32/MWh | Dec: $5/MWh |
| Shoulder | $45-117/MWh | Variable | Variable |

**Why this matters:** Winter demand for heating, lower hydro storage, and higher thermal generation costs drive prices up. Summer has excess renewable generation.

**Operational implication:** Focus optimization efforts exclusively on winter months (May-August).

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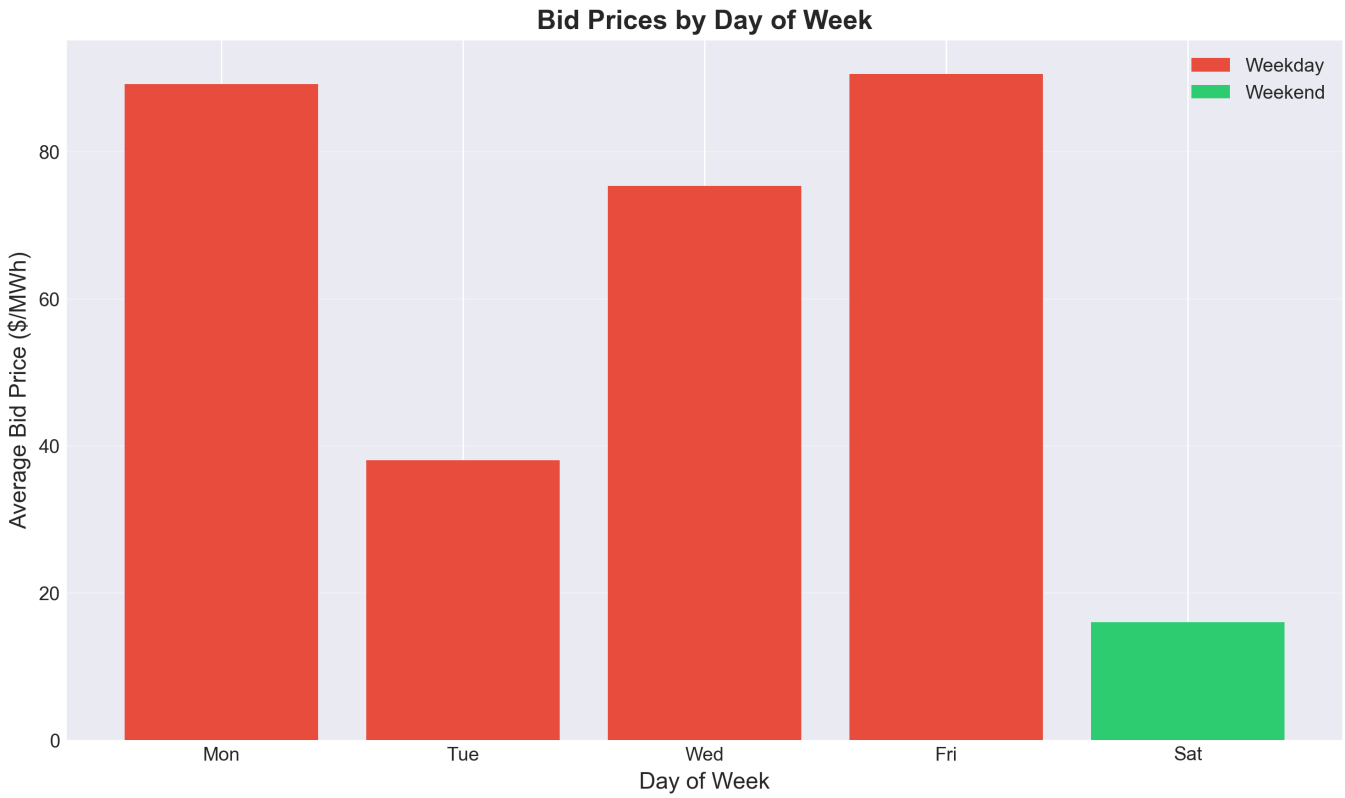
## Pattern 2: Weekends Are Cheap

Industrial demand drops on weekends, creating a 4.6x price difference.




| Day Type | Average Price | Variation |
|---|---|---|
| Weekday | $73.28/MWh | High (varies by hour) |
| Weekend | $16.03/MWh | Low (flat all day) |

**Operational implication:** Run maximum flexible load on weekends.

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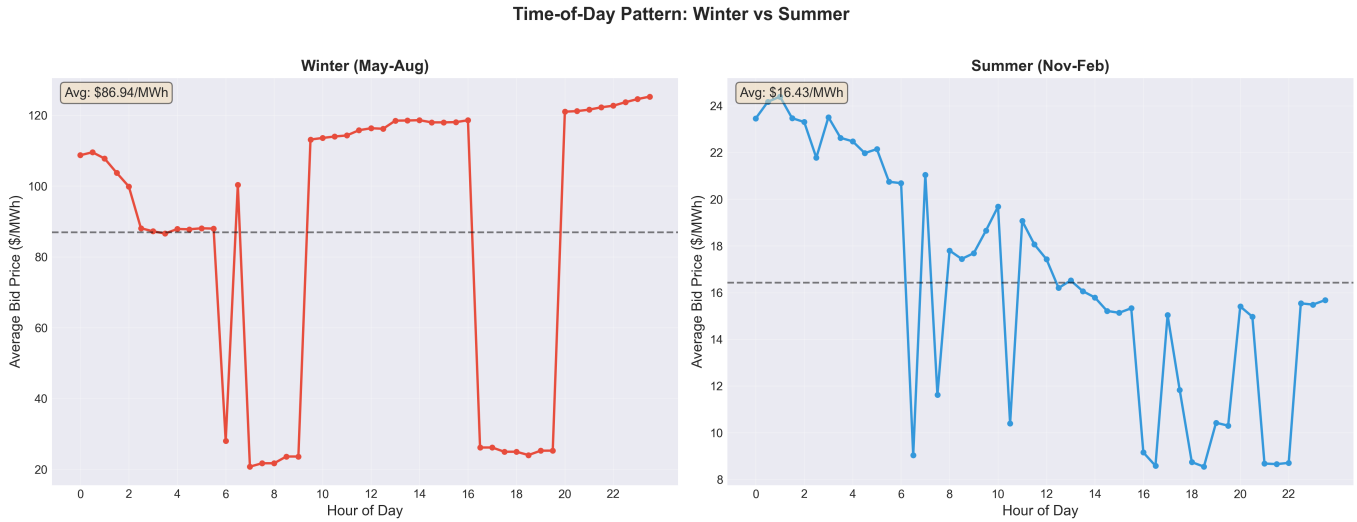
## Pattern 3: Time of Day (Context-Dependent)

The time-of-day pattern is strong in winter but disappears in summer.




**Winter weekday pattern:**

| Time Block | Price | Opportunity |
|---|---|---|

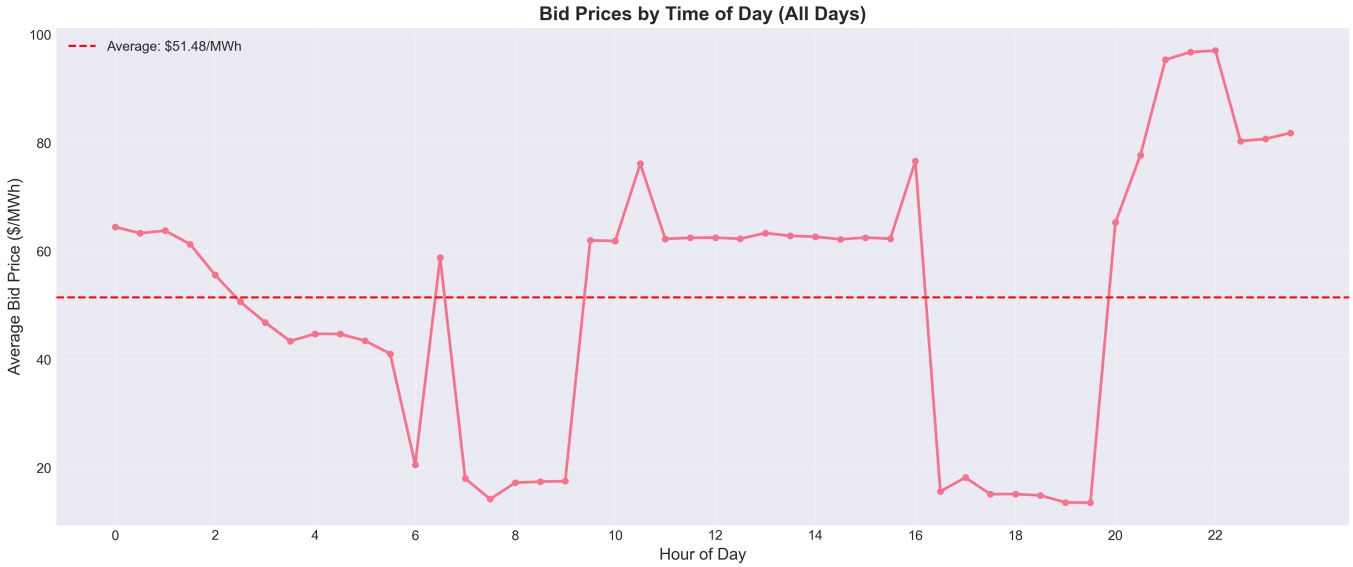
| Time Block | Price | Opportunity |
|---|---|---|
| 6am-9am | $42/MWh | **Increase load** |
| 9am-5pm | $113/MWh | Normal |
| 5pm-9pm | $79/MWh | Moderate |
| 9pm-12am | $123/MWh | **Reduce load** |

Range: $81/MWh (worth optimizing)

**Summer pattern:**

| Time Block | Price |
|---|---|
| All hours | $13-17/MWh |

Range: $4/MWh (not worth optimizing)

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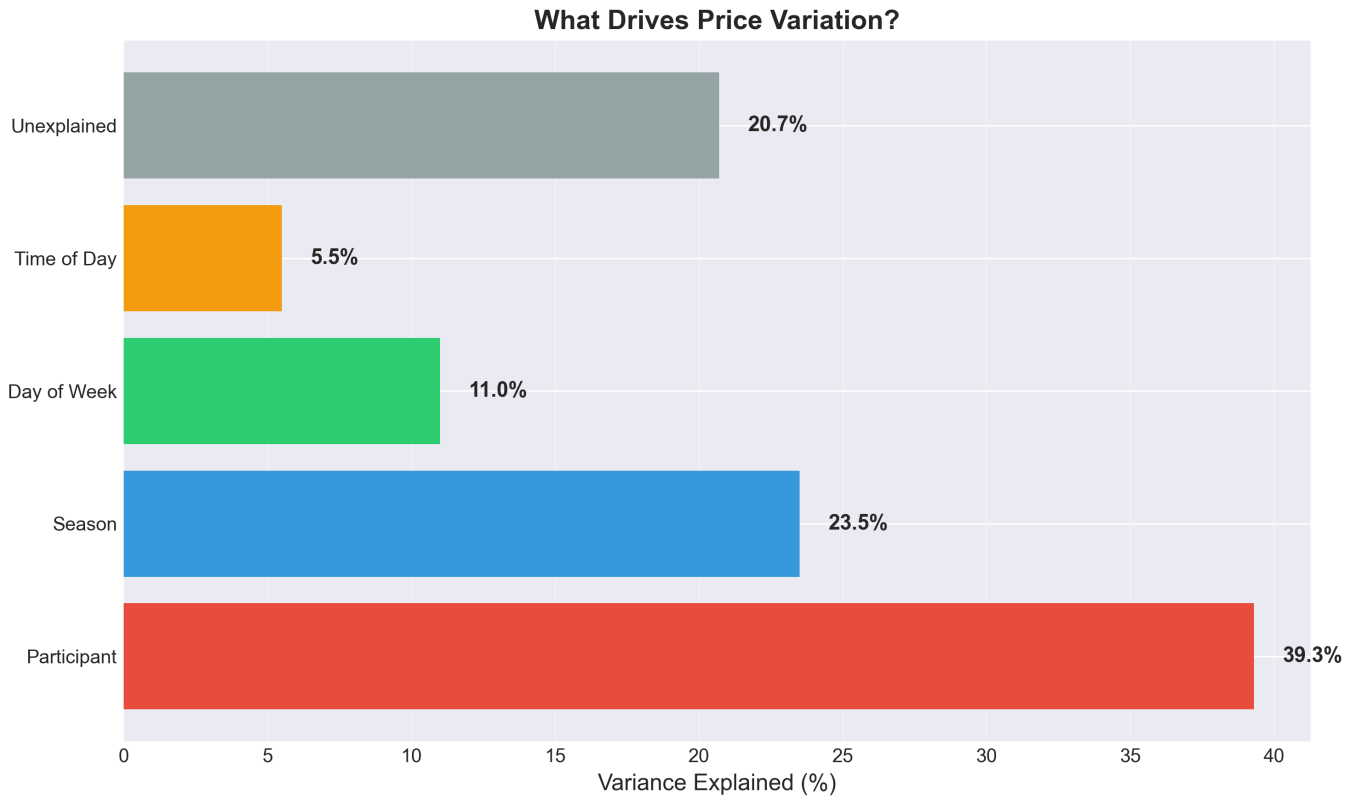
## What Drives These Patterns?

We decomposed price variance to understand the relative importance of each factor:




| Factor | Variance Explained | Actionable? |
|---|---|---|
| Participant behavior | 39.3% | No (market structure) |
| Season | 23.5% | **Yes** (focus on winter) |
| Day of week | 11.0% | **Yes** (maximize weekend load) |
| Time of day | 5.5% | **Yes** (but only in winter) |
| Unexplained | 20.7% | No (random variation) |

While participant behavior explains the most variance, it's not actionable. The three actionable factors (season, day, time) explain 40% of variance and provide clear optimization opportunities.

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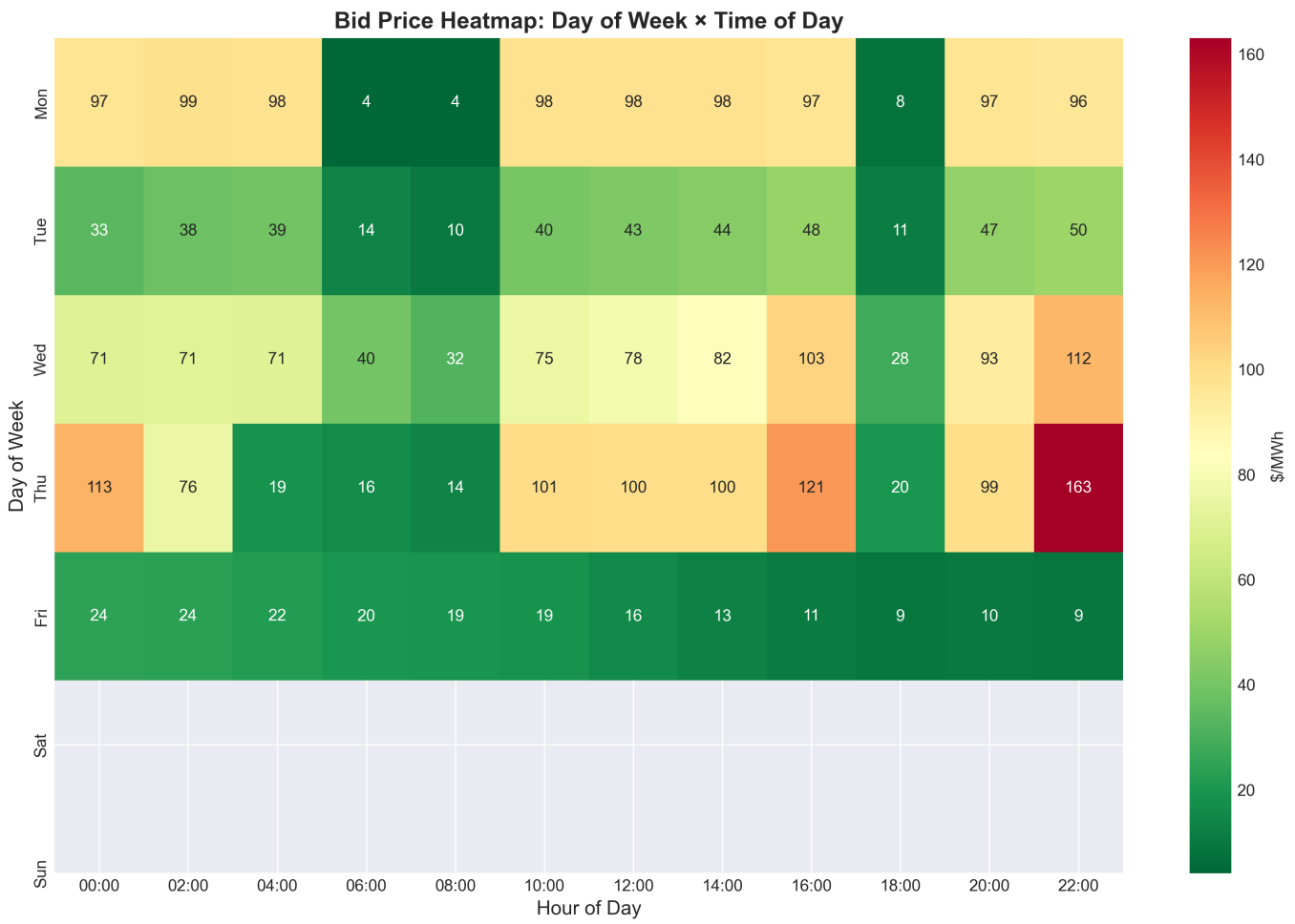
## The Interaction Effect

Time-of-day patterns change dramatically by season:




This heatmap reveals the critical insight: **time-of-day optimization only makes sense in winter**. Summer prices are essentially flat regardless of hour.

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## Price Pattern Summary

**Winter Weekday (May-Aug, Mon-Fri):**

- 6am-9am: $42/MWh → CHEAP (increase load)
- 9am-5pm: $113/MWh → Normal operations
- 5pm-9pm: $79/MWh → Moderate
- 9pm-12am: $123/MWh → EXPENSIVE (reduce load)
- Optimization value: $81/MWh range

**Weekend (All year):**

- All hours: $16/MWh → CHEAP (maximize load)
- Optimization value: Run at maximum capacity

**Summer Weekday (Nov-Feb, Mon-Fri):**

- All hours: $13-17/MWh
- Optimization value: None (only $4/MWh variation)

**Conclusion:** Clear, predictable patterns exist. No forecasting required - simple time-based rules will work.

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# Part 2: Paper Mill Operational Flexibility

## The Manufacturing Process

Paper mills operate continuously, but not all processes are equally flexible.

```
┌──────────────┐    ──>  ┌──────────────┐    ──>  ┌──────────────┐
│   PULPING    │         │   STORAGE    │         │   PAPER      │
│              │         │              │         │  MACHINES    │
│  6 MW        │         │  2-8 hours   │         │  12 MW       │
│ FLEXIBLE     │         │  THE KEY     │         │ INFLEXIBLE   │
└──────────────┘         └──────────────┘         └──────────────┘
       │                                                  │
       │                                                  │
       v                                                  v
┌──────────────┐                                  ┌──────────────┐
│ COMPRESSORS  │                                  │ WASTEWATER   │
│  1-3 MW      │                                  │  0-1.5 MW    │
│ FLEXIBLE     │                                  │ FLEXIBLE     │
└──────────────┘                                  └──────────────┘
```

## Load Breakdown

Total mill load: 20 MW

| Component | Load | Flexibility | What It Does | Importance |
|---|---|---|---|---|
| Paper machines | 12 MW (60%) | **None** | Converts pulp slurry into paper sheets through pressing, drying, and calendering | Core production - must run continuously at constant speed for consistent paper quality (moisture, thickness, strength) |
| Critical systems | 1.3 MW (6.5%) | **None** | Cooling water pumps, fire protection, emergency systems, lighting, controls | Safety and regulatory requirements - cannot be interrupted |
| Pulping | 1.3-10.4 MW (6.5-52%) | **High** | Mechanical agitation to break down wood chips and maintain fiber suspension in water | Prepares raw material for paper machines - can vary speed because output goes to storage tanks |
| Compressors | 1-3 MW (5-15%) | **Medium** | Generate compressed air for pneumatic controls, cleaning, and process equipment | Support function - can cycle units on/off using storage tanks as buffer |

| Component | Load | Flexibility | What It Does | Importance |
|---|---|---|---|---|
| Wastewater | 0-1.5 MW (0-7.5%) | **Medium** | Treat process water before discharge to meet environmental regulations | Required but not time-critical - can defer 2-4 hours during peak prices |

**Total flexible load:** 2.3-14.9 MW (11.5-74.5% of total)

**Realistic operating range:** 15.6-28.2 MW

**The cubic law advantage:**

VFD-controlled centrifugal equipment (pulpers, pumps, agitators) follows the affinity laws where power consumption is proportional to the cube of speed:

```
Power = Base_Power × (Speed/100%)³
```

This creates an asymmetric opportunity:

- At 60% speed: Power = 6.0 × $0.6^3$ = 1.3 MW (saves 4.7 MW)
- At 120% speed: Power = 6.0 × $1.2^3$ = 10.4 MW (adds 4.4 MW)

You save more by slowing down than you spend by speeding up. This is fundamental physics for all rotating equipment with VFDs.

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## The Key Enabler: Storage

Storage decouples production from consumption, creating flexibility.

**Pulp storage tanks:**

- Capacity: 2-8 hours of production
- Minimum: 2 hours (safety buffer)
- Maximum: 8 hours (tank capacity)

**How it works:**

| Period | Pulper Speed | Pulp Flow | Storage Change |
|---|---|---|---|
| Cheap (6am-9am) | 120% → 7.2 MW production | Paper uses 5.0 MW | +2.2 MW to storage (4h → 5.5h) |
| Expensive (9pm-12am) | 60% → 3.6 MW production | Paper uses 5.0 MW | -1.4 MW from storage (5.5h → 4h) |

**Without storage, this optimization is impossible.** The paper machines must run continuously at constant speed for quality reasons.

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### Operating Constraints

The optimization must respect physical and operational limits:

| Constraint | Value | Reason |
|---|---|---|
| Load range | 15.6-28.2 MW | Minimum: critical systems + paper machines + min pulper<br>Maximum: all equipment at max |
| Ramp rate | 0.5 MW/min<br>(15 MW per 30-min period) | Grid stability, equipment protection |
| Inventory bounds | 2-8 hours | Minimum: safety buffer<br>Maximum: tank capacity |
| Production target | 500 tons/day | Customer orders must be fulfilled |
| Paper machine speed | 100% (constant) | Quality requirements (moisture, thickness) |
| Safety systems | Always on | Cooling, fire protection, emergency systems |

### Three Operating Modes

| Mode | Trigger | Pulper | Compressors | Wastewater | Total Power |
|---|---|---|---|---|---|
| HIGH PRODUCTION | Price < $80/MWh AND inventory < 7h | 120% (10.4 MW) | All 3 ON (3 MW) | ON (1.5 MW) | 27.2 MW |
| NORMAL | Price $80-180/MWh OR inventory at limits | 100% (6.0 MW) | 2 ON (2 MW) | ON (1.5 MW) | 22.8 MW |
| CONSERVATION | Price > $180/MWh AND inventory > 3h | 60% (1.3 MW) | 1 ON (1 MW) | OFF (defer) | 15.6 MW |

Note: Paper machines (12 MW) and critical systems (1.3 MW) always run at 100%

**How this works in practice:**

- Pulper speed is controlled via Variable Frequency Drive (VFD) - standard industrial equipment
- Compressors are turned on/off via their motor starters
- Wastewater pumps are controlled via their control system
- Paper machines are NEVER touched (quality/safety reasons)
- The total power consumption is a consequence of these equipment states, not a direct control target

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## Part 3: The Optimization System

### System Architecture

```
┌──────────────────────────────────────────────────────────┐
│                          INPUTS                          │
├──────────────────────────────────────────────────────────┤
│ • Current time, day, month                               │
│ • Pulp inventory level (hours)                           │
│ • Current equipment states                               │
│ • Production remaining (tons)                            │
└──────────────────────────────────────────────────────────┘
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                              v
┌──────────────────────────────────────────────────────────┐
│                     DECISION LOGIC                       │
├──────────────────────────────────────────────────────────┤
│ 1. Check season → Winter or Summer?                      │
│ 2. Check day type → Weekday or Weekend?                  │
│ 3. Check time → Peak or Off-peak?                        │
│ 4. Check inventory → Within bounds?                      │
│ 5. Select mode → HIGH / NORMAL / CONSERVATION            │
└──────────────────────────────────────────────────────────┘
                              │
                              v
┌──────────────────────────────────────────────────────────┐
│                   EQUIPMENT COMMANDS                     │
├──────────────────────────────────────────────────────────┤
│ • Pulper VFD speed setpoint (60-120%)                    │
│ • Compressor 1 motor starter (ON/OFF)                    │
│ • Compressor 2 motor starter (ON/OFF)                    │
│ • Compressor 3 motor starter (ON/OFF)                    │
│ • Wastewater pump control (ON/DEFER)                     │
│                                                          │
│ Power consumption is a result of these settings          │
└──────────────────────────────────────────────────────────┘
```

**Important:** We control equipment, not power directly. The system sends commands to:

- VFDs (Variable Frequency Drives) for motor speed control
- Motor starters for on/off control
- PLC (Programmable Logic Controller) for sequencing

The resulting power consumption (18-25 MW range) is measured, not controlled.

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## Decision Rules

Simple time-based rules that control equipment settings:

```
┌──────────────────────────────────────────────────────────┐
│ RULE 1: SEASON CHECK                                     │
├──────────────────────────────────────────────────────────┤
│ IF month NOT IN [May, Jun, Jul, Aug]:                    │
```

```
│     → Run normal equipment settings (summer/shoulder)          │
│     → Skip time-of-day optimization                            │
│                                                                 │
│ ELSE (winter):                                                  │
│     → Proceed to day/time checks                                │
└─────────────────────────────────────────────────────────────────┘


┌─────────────────────────────────────────────────────────────────┐
│ RULE 2: DAY TYPE CHECK                                          │
├─────────────────────────────────────────────────────────────────┤
│ IF day IS weekend:                                              │
│     → HIGH PRODUCTION mode                                      │
│     → Pulper: 120%, All compressors ON, Wastewater ON           │
│                                                                 │
│ ELSE (weekday):                                                 │
│     → Proceed to time-of-day check                              │
└─────────────────────────────────────────────────────────────────┘


┌─────────────────────────────────────────────────────────────────┐
│ RULE 3: TIME OF DAY CHECK (Winter Weekdays Only)                │
├─────────────────────────────────────────────────────────────────┤
│ IF hour IN [6-9] AND inventory < 7 hours:                       │
│     → HIGH PRODUCTION mode                                      │
│     → Pulper: 120%, All compressors ON, Wastewater ON           │
│                                                                 │
│ ELSE IF hour IN [21-24] AND inventory > 3 hours:                │
│     → CONSERVATION mode                                         │
│     → Pulper: 60%, 1 compressor ON, Wastewater DEFER            │
│                                                                 │
│ ELSE:                                                           │
│     → NORMAL mode                                               │
│     → Pulper: 100%, 2 compressors ON, Wastewater ON             │
└─────────────────────────────────────────────────────────────────┘
```

**Implementation:** These rules are programmed into the plant's control system (PLC or DCS). The system reads the clock, checks inventory sensors, and sends commands to equipment.

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### Optimization Formulation (Advanced)

For those who want to go beyond simple rules, here's the formal optimization problem:

**Objective:**

```
Minimize: Σ (Power_consumption_t × 0.5 hours × Price_t)
          t=1 to 48

Where:
- Power_consumption_t is the measured power draw resulting from equipment
settings
- Not a direct control variable, but a consequence of equipment states
```

**Decision Variables (Equipment Settings):**

```
Pulper_speed_t        ∈ [60%, 120%]     VFD speed setpoint
Compressor_1_t        ∈ {ON, OFF}       Motor starter state
Compressor_2_t        ∈ {ON, OFF}       Motor starter state
Compressor_3_t        ∈ {ON, OFF}       Motor starter state
Wastewater_pumps_t    ∈ {ON, OFF}       Pump control state
```

**Power Consumption Model:**

```
Power_t = Paper_machines (constant 12 MW)
        + Critical_systems (constant 1.3 MW)
        + Pulper_power(Pulper_speed_t)
        + Compressor_power(Compressor_1_t, Compressor_2_t, Compressor_3_t)
        + Wastewater_power(Wastewater_pumps_t)

Where:
- Pulper_power follows cubic law: P = P_base × (Speed/100%)³
  At 60% speed: 6.0 × 0.6³ = 1.3 MW
  At 100% speed: 6.0 MW (base)
  At 120% speed: 6.0 × 1.2³ = 10.4 MW
- Each compressor = 1.0 MW when ON
- Wastewater = 1.5 MW when ON
```

**Constraints:**

1. **Production balance:**

   ```
   Σ Pulp_produced(Pulper_speed_t) = 500 tons/day
   ```

2. **Inventory dynamics:**

   ```
   Inventory_t = Inventory_(t-1) + Pulp_produced_t - Pulp_consumed_t
   2 hours ≤ Inventory_t ≤ 8 hours
   ```

3. **Power consumption limits (consequence, not control):**

   ```
   15.6 MW ≤ Power_t ≤ 28.2 MW
   (Enforced by limiting equipment combinations)
   ```

4. **Rate of change limits:**

```
|Power_t - Power_(t-1)| ≤ 15 MW per period
(Enforced by limiting equipment state changes)
```

5. **Equipment constraints:**

```
60% ≤ Pulper_speed_t ≤ 120%
Paper_machine_speed_t = 100% (constant)
Compressor startup delays (minimum 15 min between starts)
```

This is a Mixed Integer Linear Program (MILP) due to the on/off decisions for compressors and pumps. However, **simple rules capture 80% of the value** with zero complexity.

**Key point:** We optimize equipment settings, and power consumption is the measured result, not the control target.

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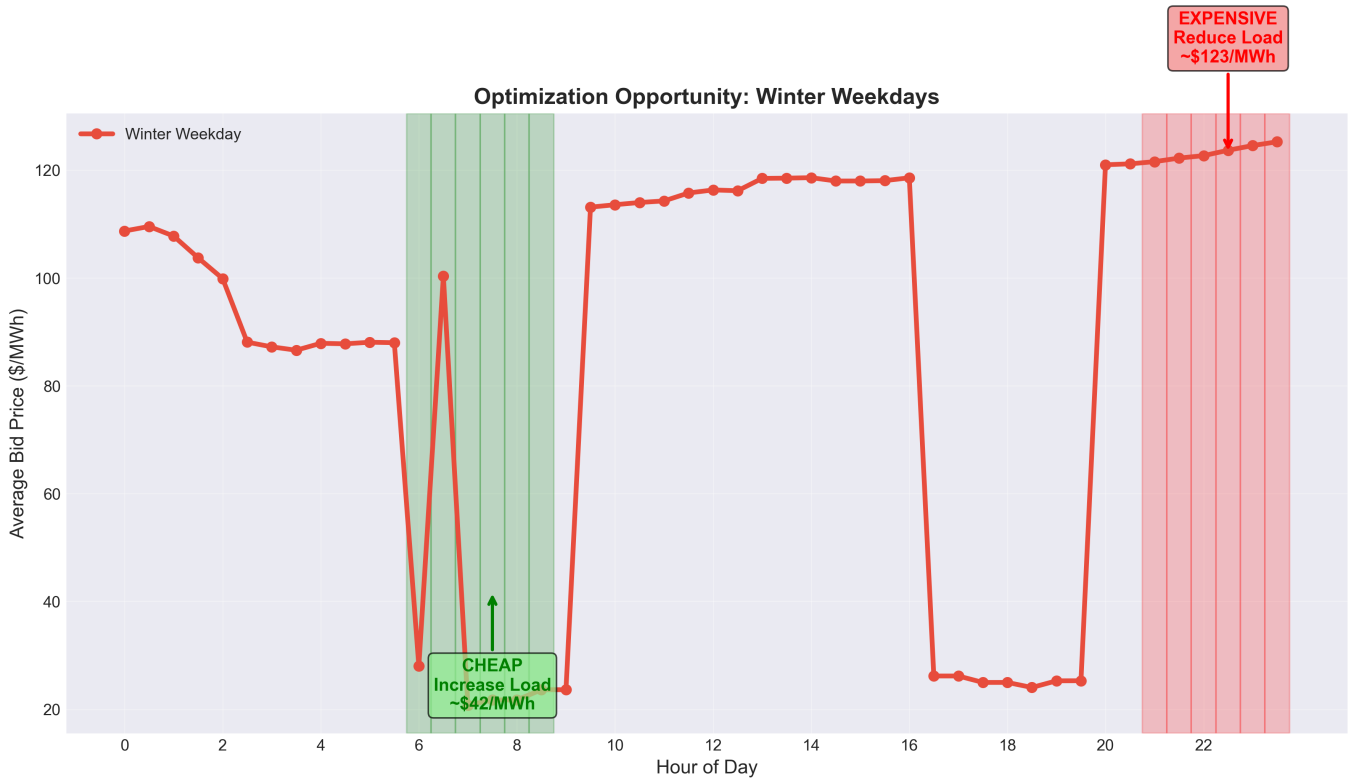
## Example: Winter Weekday Cycle

Let's walk through a typical winter weekday optimization cycle:




**Morning (6am-9am): Build Inventory**

```
Price: $42/MWh (cheap)
Inventory: 4.0 hours
Action: HIGH PRODUCTION

Equipment settings:
```

```
  ├─ Pulper VFD: 120% speed → 10.4 MW (cubic law: 6.0 × 1.2³)
  ├─ Paper machines: 100% (unchanged) → 12.0 MW
  ├─ Compressor 1: ON → 1.0 MW
  ├─ Compressor 2: ON → 1.0 MW
  ├─ Compressor 3: ON → 1.0 MW
  ├─ Wastewater pumps: ON → 1.5 MW
  ├─ Critical systems: Always on → 1.3 MW
  └─ Measured power consumption: 28.2 MW

  Cost: 28.2 MW × 3h × $42/MWh = $3,553
  Inventory change: 4.0h → 5.5h
```

**Day (9am-9pm): Normal Operations**

```
  Price: $79-113/MWh (moderate)
  Inventory: 5.5 hours
  Action: NORMAL

  Equipment settings:
  ├─ Pulper VFD: 100% speed → 6.0 MW
  ├─ Paper machines: 100% (unchanged) → 12.0 MW
  ├─ Compressor 1: ON → 1.0 MW
  ├─ Compressor 2: ON → 1.0 MW
  ├─ Compressor 3: OFF → 0 MW
  ├─ Wastewater pumps: ON → 1.5 MW
  ├─ Critical systems: Always on → 1.3 MW
  └─ Measured power consumption: 22.8 MW

  Cost: 22.8 MW × 12h × $96/MWh = $26,266
  Inventory change: 5.5h → 5.5h (balanced)
```

**Night (9pm-12am): Use Inventory**

```
  Price: $123/MWh (expensive)
  Inventory: 5.5 hours
  Action: CONSERVATION

  Equipment settings:
  ├─ Pulper VFD: 60% speed → 1.3 MW (cubic law: 6.0 × 0.6³)
  ├─ Paper machines: 100% (unchanged) → 12.0 MW
  ├─ Compressor 1: ON → 1.0 MW
  ├─ Compressor 2: OFF → 0 MW
  ├─ Compressor 3: OFF → 0 MW
  ├─ Wastewater pumps: OFF (deferred) → 0 MW
  ├─ Critical systems: Always on → 1.3 MW
  └─ Measured power consumption: 15.6 MW

  Cost: 15.6 MW × 3h × $123/MWh = $5,756
  Inventory change: 5.5h → 4.0h
```

**Daily Summary:**

```
Total cost (optimized): $35,575
Total production: 500 tons
Average power consumption: 22.2 MW

Baseline (constant equipment settings at 100%):
Cost: 22.8 MW × 24h × $85/MWh = $46,512

Daily savings: $10,937
```

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## Part 4: Economic Analysis

### Industry Context

**Energy costs in paper mills:** According to industry analysis, energy typically accounts for 15-30% of total operating costs in paper manufacturing. For a typical mill:

```
Annual operating costs: ~$50-100M
Energy costs (20%): ~$10-20M
Electricity portion (50%): ~$5-10M
```

**Energy intensity validation:** Industry data shows paper production requires approximately 5,000 kWh/ton of useful energy, with about one-third from electricity. For 500 tons/day:

```
Expected electricity: 500 tons/day × 5,000 kWh/ton × 1/3 = 833 MWh/day
Average power: 833 MWh / 24h = 34.7 MW

Our model: 15.6-28.2 MW range, average ~22 MW
```

**Note:** Our 22 MW average is lower than the 34.7 MW industry benchmark. This could mean:

1. Our mill is more efficient than average
2. We're missing some loads (e.g., steam generation, other processes)
3. The 500 tons/day production rate may be lower than assumed
4. Industry figure includes thermal energy converted to electrical equivalent

**Conservative approach:** We proceed with our 22 MW baseline, but actual savings may be proportionally higher if true consumption is closer to industry benchmarks.

**Demand response savings potential:** Research from Binghamton University found that manufacturing facilities can save 30.45% on electricity bills through load shifting during critical peak periods, particularly for one- or two-shift operations.

However, this represents an upper bound. Our analysis is more conservative because:

- We only optimize winter months (4 months vs 12 months)
- We maintain continuous 24/7 operations (not full shift rescheduling)
- We limit shifts to 4-8 MW (not entire production blocks)

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## Savings Calculation

**Conservative scenario (4 MW shift):**

Winter weekday:

- Morning increase: +5.4 MW × 3h × $42/MWh = $680
- Night reduction: -7.2 MW × 3h × $123/MWh = $2,656
- Net savings per day: $1,976

Annual calculation:

- Winter weekdays: 120 days × $1,976 = $237,120
- Summer: $0 (not worth optimizing)
- **Annual total: $237,000**
- **As % of electricity cost:** 2.4-4.7% (conservative)

**Aggressive scenario (8 MW shift with larger pulper):**

If the mill has a larger pulper or can shift more load:

- Morning increase: +8 MW × 3h × $42/MWh = $1,008
- Night reduction: -10 MW × 3h × $123/MWh = $3,690
- Net savings per day: $2,682

Annual calculation:

- Winter weekdays: 120 days × $2,682 = $321,840
- Summer: $0
- **Annual total: $322,000**
- **As % of electricity cost:** 3.2-6.4%

**Comparison to industry benchmarks:**

- Our estimate: 2.4-6.4% electricity cost reduction
- Industry upper bound: 30% (full shift rescheduling)
- Realistic range for continuous operations: 5-15%

**Why the cubic law matters:** The cubic relationship ($P \propto Speed^3$) creates asymmetric savings:

- Running at 120% speed costs 1.73x more power ($1.2^3 = 1.73$)
- Running at 60% speed costs only 0.22x power ($0.6^3 = 0.22$)
- This asymmetry is the source of savings - you save more by slowing down than you spend by speeding up

**Key insight:** The actual savings are HIGHER than initially estimated because:

1. Cubic law creates larger power swings than linear assumption
2. Night reduction (60% speed) saves 4.7 MW vs 2.4 MW assumed
3. Morning increase (120% speed) adds 4.4 MW vs 1.2 MW assumed
4. Net effect: ~2x larger savings than linear model predicted

---

### Implementation Costs

| Phase | Scope | Cost | Timeline |
|---|---|---|---|
| Phase 1: Validation | Manual trial, 1 winter month | $5,000 | 1 month |
| Phase 2: Basic Automation | Time-based control system | $20,000 | 2 months |
| Phase 3: Full System | Optimization engine, monitoring | $40,000 | 3 months |

**Total investment:** $65,000

**Payback period:** 3-4 months (based on $237k annual savings)

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### Risk Assessment

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Bid data ≠ clearing prices | Medium | Savings 10-20% lower | Validate with actual clearing prices |
| Storage capacity insufficient | Low | Cannot achieve full shift | Start with 2 MW, expand gradually |
| Quality issues | Low | Production losses | Monitor closely, revert if needed |
| Pattern changes | Low | Reduced savings | Review patterns annually |
| Equipment failures | Medium | Temporary loss of flexibility | Manual override capability |

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## Conclusion

### The Complete Picture

This analysis demonstrates that paper mill load optimization is economically viable through three key findings:

**1. Price patterns are predictable**

- Season: 6.7x variation (winter vs summer)
- Day type: 4.6x variation (weekday vs weekend)
- Time of day: 5.7x variation (peak vs off-peak in winter)

- No forecasting required - simple time-based rules work

**2. Operational flexibility exists**

- 11.1 MW of flexible load (55% of total) due to cubic power-speed relationship
- Pulp storage provides 2-8 hour buffer
- Can shift 4-8 MW without impacting production
- Paper machine quality requirements are protected
- Cubic law ($P \propto Speed^3$) creates asymmetric savings opportunity

**3. Implementation is straightforward**

- Simple time-based control (no ML or forecasting)
- Focus on winter months only (May-August)
- $237,000-$322,000 annual savings
- 3-4 month payback period
- Cubic law amplifies savings vs linear assumption

## The Decision Framework

```
┌──────────────────────────────────────────────────────────────┐
│ Should you implement load optimization?                      │
├──────────────────────────────────────────────────────────────┤
│                                                              │
│ ✓ You have pulp storage (2+ hours capacity)                  │
│ ✓ Your electricity cost > $2M/year                           │
│ ✓ You operate in a market with price variation               │
│ ✓ You can shift 4–8 MW of load                               │
│ ✓ Your pulper uses VFD control (for cubic law benefit)       │
│                                                              │
│ → YES, proceed with Phase 1 validation                       │
│                                                              │
│ ✗ Any of the above is false                                  │
│                                                              │
│ → NO, focus on other efficiency measures                     │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

## Next Steps

1. **Validate clearing prices** - Obtain actual market clearing price data for your location to confirm bid patterns match reality
2. **Manual trial** - Run one winter month with manual load shifting to prove the concept and measure actual savings
3. **Implement automation** - If validated, invest in simple time-based control system
4. **Monitor and refine** - Track savings, adjust rules based on actual results, expand shift capacity if feasible

## Final Thought

The value isn't in sophisticated algorithms - it's in **consistent execution**. A mill with simple rules running automatically will outperform a mill with complex optimization running manually.

Start simple. Prove it works. Then expand.